

Experiment 'p_2_10_hybrid_car_cost' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_10_hybrid_car_cost

Natural language statement: Write a method that helps a person decide whether to buy a hybrid car. Your methods's inputs should be: the cost of a new car, the estimated miles driven per year, the estimated gas price, the efficiency in miles per gallon, the estimated resale value after 5 years. Compute the total cost of owning the car for five years.

Method signature: p_2_10_hybrid_car_cost(new_car_cost: real, miles_per_year: real, gas_price: real, mpg: real, resale_value: real) returns (total_cost: real)

Ensures

- `total_cost == new_car_cost + (miles_per_year / mpg) * gas_price * 5.0 - resale_value`

Requires

- `0.0 < mpg`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that helps a person decide whether to buy a hybrid car. Your methods's
→ inputs should be: the cost of a new car, the estimated miles driven per year, the
→ estimated gas price, the efficiency in miles per gallon, the estimated resale
→ value after 5 years. Compute the total cost of owning the car for five years.

The signature should be:

`p_2_10_hybrid_car_cost(new_car_cost: real, miles_per_year: real, gas_price: real, mpg:
→ real, resale_value: real) returns (total_cost: real)`

The method should respect the following contract:

`requires 0.0 < mpg, ensures total_cost == new_car_cost + (miles_per_year / mpg) *
→ gas_price * 5.0 - resale_value`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdhuHQBSzgse0HJWxh7yFcHhL4SF

Model: gpt-5.5-2026-04-23

Created at: 1780484750

Finish reason: stop

Usage: 470 tokens in, and 43 tokens out

```
{  
  total_cost := new_car_cost + (miles_per_year / mpg) * gas_price * 5.0 - resale_value;  
}
```

Final Program

```
method p_2_10_hybrid_car_cost(new_car_cost: real, miles_per_year: real, gas_price: real,  
→ mpg: real, resale_value: real) returns (total_cost: real)  
  requires 0.0 < mpg  
  ensures total_cost == new_car_cost + (miles_per_year / mpg) * gas_price * 5.0 -  
→ resale_value  
{  
  total_cost := new_car_cost + (miles_per_year / mpg) * gas_price * 5.0 - resale_value;  
}
```

Total Token Usage

Input tokens: 470

Output tokens: 43

Reasoning tokens: 0

Sum of 'total tokens': 513

Experiment Timings

Overall Experiment started at 1780484749177, ended at 1780484756573, lasting 7396ms (7.40 seconds)

Iteration #1 started at 1780484749178, ended at 1780484756573, lasting 7395ms (7.40 seconds)